

ADVANCE PROBLEM PROGRAMMING

SLO 2: Apply Control Structures

1. Prime Number

A prime number is a number greater than 1 that has only two factors: 1 and itself.

```
public class Main {  
    public static void main(String[] args) {  
        int n = 7, flag = 0;  
  
        for (int i = 2; i <= n / 2; ++i) {  
            if (n % i == 0) {  
flag = 1;  
break;  
            }  
        }  
  
        System.out.println(flag == 0 ? "Prime" : "Not Prime");  
    }  
}
```

Output:

Prime

2. Factorial

The factorial of a number n is the product of all positive integers from 1 to n.

```
public class Main {  
    public static void main(String[] args) {  
        int n = 5;  
        int fact = 1;  
  
        for (int i = 1; i <= n; i++) {  
fact *= i;  
        }  
  
        System.out.println("Factorial = " + fact);  
    }  
}
```

Output:

Factorial = 120

3. Largest of Three Numbers

The following program uses if-else selection to find the largest among three numbers.

```
public class Main {  
    public static void main(String[] args) {  
        int a = 25, b = 15, c = 20;  
  
        if (a > b && a > c) {  
System.out.println("Largest = " + a);  
        } else if (b > c) {  
System.out.println("Largest = " + b);  
        } else {  
System.out.println("Largest = " + c);  
        }  
    }  
}
```

Output:

Largest = 25